

EdgeMind

A private, on-device AI workspace copilot optimised for
Snapdragon-powered HP PCs

Project Proposal

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Prototype: github.com/ayush2459/edgemind-snapdragon

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Note on figures: performance and battery numbers in this document are design targets to be validated on a Snapdragon X HP PC. They are not measured results.

1. Executive summary

Knowledge workers keep their most valuable information in documents, notes and meetings, yet the AI tools that could unlock it usually require sending that information to the cloud. That creates privacy risk, recurring cost, latency and a dependence on connectivity.

EdgeMind is a private AI workspace copilot that runs entirely on the laptop. It indexes a user's files and meeting audio locally, then answers questions, summarises long material and searches across everything with cited sources. It is designed from the ground up for **Snapdragon-powered HP PCs**: models are executed on the Hexagon NPU through ONNX Runtime's QNN execution provider, the application is built ARM64-native, and a power-aware policy engine adapts NPU behaviour to battery state.

Key points

- **Private by design:** no document text leaves the device, which suits regulated industries.
- **Optimised for the platform:** NPU-first inference, quantised models, compiled-graph caching and ARM64-native code.
- **Battery-aware:** the policy engine trades speed for efficiency when unplugged.
- **Works offline:** full functionality without a network connection.
- **Working prototype:** a tested reference implementation is included in the accompanying Git repository.

2. Problem statement

Cloud-hosted AI assistants have made summarisation and question answering routine, but several barriers keep them out of many workplaces:

Barrier	Impact
Data privacy and compliance	Legal, healthcare, finance and government teams often cannot upload documents to third-party services.
Recurring cost	Per-seat or per-token pricing scales with usage and becomes hard to budget for large teams.
Latency and connectivity	Round trips to a server add delay and fail entirely when the network is poor or absent.
Battery and thermals	Running AI on the CPU drains the battery and heats the device, which discourages on-device use.
Fragmented information	Answers are spread across files, chats and meetings, with no single private place to search them.

The arrival of laptops with dedicated NPUs changes what is possible locally. What is missing is an application designed around that hardware rather than a cloud tool ported to it.

3. Proposed solution

EdgeMind is a local-first AI copilot with four core capabilities, all running on the user's Snapdragon-powered HP PC.

Capability	Description	Primary compute
Private document Q&A	Retrieval-augmented answers across PDFs, notes and text files, with source citations.	NPU (embeddings), NPU/CPU (LLM)
Meeting transcription and notes	Speech-to-text with searchable transcripts and generated summaries and action items.	NPU (speech model)
Summarise anything	Condense long documents or folders into short briefs at a chosen length.	NPU/CPU
Unified semantic search	One search box across documents and transcripts, ranked by meaning rather than keywords.	NPU (embeddings)

User journey

- The user points EdgeMind at chosen folders; files are chunked and embedded locally in the background at low power.
- They ask a question in natural language. The retrieval service finds the most relevant passages.
- The language service composes an answer grounded in those passages and lists the source files.
- Meeting audio is transcribed on-device, indexed, and becomes searchable alongside documents.

4. Why Snapdragon-powered HP PCs

The project is intentionally built around the strengths of this platform rather than being generic software that happens to run on it.

Platform strength	Relevance to EdgeMind
Hexagon NPU (up to 45 TOPS on Snapdragon X series)	Sustained inference for embeddings and speech at low power, meeting the class of hardware behind Copilot+ PCs.
Oryon ARM64 CPU cores	Efficient orchestration, parsing and indexing without x64 emulation overhead when built ARM64-native.
Adreno GPU	Optional acceleration path through DirectML for models that suit the GPU.
Long battery life	Enables all-day background indexing and meeting capture that would be impractical on a power-hungry CPU.
Unified memory (LPDDR5X)	Fast data movement between CPU, GPU and NPU for retrieval workloads.
HP business PC line and HP Wolf Security	Fits enterprise deployment, manageability and hardware-backed security expectations.

5. System architecture

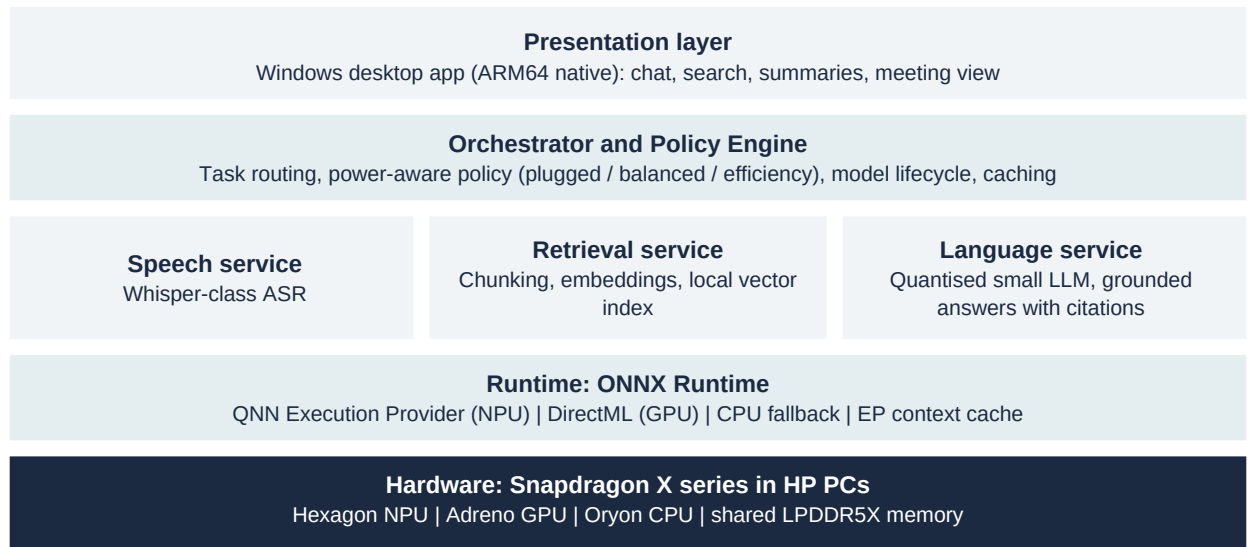


Figure 1. EdgeMind layered architecture. All layers run on the device; there is no cloud dependency.

Component responsibilities

Component	Responsibility
Presentation layer	ARM64-native desktop UI for chat, search, summaries and meeting views.
Orchestrator and policy engine	Routes tasks to services, selects the execution profile from battery state, manages model loading and caches.
Speech service	Streams audio to a quantised Whisper-class model and produces timestamped transcripts.
Retrieval service	Chunks text, generates embeddings, stores vectors in a local index and returns top-k passages.
Language service	Runs a small quantised LLM for grounded answers and summaries, with extractive fallback.
Runtime	ONNX Runtime with the QNN execution provider first, then DirectML, then CPU.

6. Snapdragon optimisation strategy

Optimisation is the central requirement of this proposal. The following techniques are applied across the stack.

Technique	What we do	Benefit
NPU-first execution	Run models through the QNN execution provider (HTP backend) with ordered fallback to DirectML and CPU.	Low-power, high-throughput inference; CPU stays free.
Quantisation	Use INT8 and 4-bit weight-quantised models with static input shapes suitable for the NPU.	Smaller memory footprint, faster NPU execution.

Technique	What we do	Benefit
Compiled-graph caching	Enable EP context caching so compiled NPU graphs are stored and reused.	Faster launches after the first run.
Power-aware policy engine	Switch NPU performance mode and batch size by battery level and plug state.	Extends unplugged runtime; full speed when plugged in.
ARM64-native build	Ship native ARM64 binaries and dependencies, avoiding emulation.	Better CPU efficiency and responsiveness.
Incremental background indexing	Index only changed files in small batches at efficient power modes.	Low battery impact, always-fresh index.
Heterogeneous scheduling	Assign speech and embeddings to NPU, orchestration to CPU, optional workloads to GPU.	Parallelism without contention.
Memory discipline	Lazy-load models, share tokenizers, keep the vector index in a compact float32 matrix.	Runs comfortably alongside everyday apps.

Power policy in the prototype

Device state	HTP performance mode	Batch size
Plugged in or no battery	burst	32
On battery, above 50%	balanced	16
On battery, 20% to 50%	high_power_saver	8
On battery, below 20%	power_saver	4

7. Technology stack

Layer	Choice	Reason
Language	Python for the core; C# or Rust/Tauri for the desktop shell	Rapid development plus native ARM64 UI options.
Inference runtime	ONNX Runtime with QNN EP (onnxruntime-qnn)	Official path to the Snapdragon NPU.
Embedding model	Quantised MiniLM or BGE-small class (ONNX)	Small, accurate and NPU-friendly.
Speech model	Whisper-class model, quantised	Strong accuracy for meeting audio.
Language model	4-bit small LLM via onnxruntime-genai	Grounded answers within laptop memory limits.
Vector store	Local NumPy matrix with JSON metadata; upgrade path to SQLite or FAISS	Simple, portable and dependency-light.
Packaging	MSIX, deployable through Intune	Enterprise-friendly distribution.
Testing	pytest, on-device benchmark harness	Repeatable validation.

8. Privacy, security and compliance

- **Local-only processing:** documents, embeddings, transcripts and prompts never leave the device.
- **Encryption at rest:** the index can be stored on a BitLocker-protected drive backed by the TPM.
- **No telemetry by default:** any usage analytics would be opt-in and content-free.
- **Transparent answers:** every answer lists source files so users can verify claims.
- **Access control:** the index inherits operating-system file permissions and per-user profiles.
- **Enterprise fit:** complements HP business security features and supports policy-managed deployment.

Because no content is transmitted, EdgeMind reduces the surface for data-residency and confidentiality concerns. Final compliance mapping for a specific organisation (for example GDPR or sector rules) should be confirmed with its own legal and security teams.

9. Implementation roadmap

Phase	Timeline	Deliverables
1. Foundation	Weeks 1 to 2	Repository, hardware/provider detection, policy engine, vector store, CLI prototype with tests (completed in this proposal's prototype).
2. NPU validation	Weeks 3 to 4	Run the ONNX embedder on a Snapdragon X HP PC through QNN EP; record CPU vs NPU benchmarks; enable context caching.
3. Speech and LLM	Weeks 5 to 7	Integrate quantised speech model for meetings and a small local LLM for grounded answers with citations.
4. Desktop application	Weeks 8 to 10	ARM64-native UI, background indexer, settings, battery-aware behaviour surfaced to the user.
5. Hardening and pilot	Weeks 11 to 12	Battery and thermal testing, security review, MSIX packaging, pilot with a small user group.

10. Performance targets and validation plan

The following are design targets. Each will be measured on a Snapdragon X HP PC and compared against a CPU-only baseline on the same machine.

Metric	Target	How it is validated
Embedding throughput (NPU vs CPU)	Meaningfully higher on NPU with lower CPU utilisation	Benchmark harness (bench command), CPU utilisation counters.
Question-to-answer latency	Interactive, a few seconds for grounded answers	Timed end-to-end runs on a fixed document set.
Battery impact of background indexing	Minimal drain versus idle	Controlled unplugged runs with battery reports.
Cold start with context cache	Noticeably faster than first launch	Compare first and subsequent launches.

Metric	Target	How it is validated
Retrieval quality	Relevant source in top 4 for the large majority of test questions	Labelled question set with hit-rate scoring.
Transcription accuracy	Competitive word error rate on meeting audio	Standard test recordings.

Exact numeric thresholds will be fixed once baseline measurements exist, so that targets are grounded in real hardware rather than assumptions.

11. Impact and use cases

Audience	Use case	Value
Legal and compliance teams	Search and summarise case files and policies.	Confidential documents stay on company hardware.
Healthcare administrators	Query internal guidelines and notes.	Supports data-handling obligations.
Consultants and analysts	Ask questions across research and client material.	Faster synthesis without cloud exposure.
Students and researchers	Study from papers and lecture recordings offline.	Works without internet, extends battery life.
Field and travelling staff	Use AI on flights or in low-connectivity sites.	Full function with no network.

Expected benefits

- Lower cost per user than metered cloud AI, since inference uses hardware the organisation already owns.
- Reduced data-leak risk and simpler security reviews.
- Showcases the practical value of NPU-equipped Snapdragon HP PCs with an everyday productivity workload.

12. Feasibility, risks and mitigation

Feasibility is supported by mature components: ONNX Runtime with the QNN provider, quantised open models and a working prototype of the orchestration layer.

Risk	Likelihood / impact	Mitigation
Some operators are unsupported on the NPU	Medium / Medium	Ordered provider fallback to GPU or CPU; choose NPU-friendly model architectures.
Small LLM answer quality is limited	Medium / Medium	Strong retrieval, citations, extractive fallback, prompt constraints to context only.

Risk	Likelihood / impact	Mitigation
Model conversion and quantisation effort	Medium / Medium	Start from published pre-optimised models; automate conversion in a script.
Memory pressure on 16 GB devices	Low / Medium	Lazy loading, model unloading when idle, compact index.
Toolchain or API changes (for example genai)	Medium / Low	Isolate behind an interface; pin versions; keep fallbacks.
Access to hardware for benchmarking	Medium / High	Develop on the CPU fallback path now; reserve device time for validation phase.

13. Deployment and go-to-market

- **Distribution:** MSIX package for direct install and Intune-managed enterprise rollout.
- **Editions:** a free personal edition, and a managed enterprise edition with policy controls (allowed folders, model set, update channel).
- **Pilot first:** partner with a privacy-sensitive team such as legal or finance to gather real feedback.
- **Channel fit:** position as a reference workload for Snapdragon-powered HP PCs, highlighting battery life and privacy.

14. Future scope

- Multilingual transcription and retrieval, including Indian languages.
- Email and calendar connectors that index locally through user-granted access.
- Vision models for on-device document and screenshot understanding.
- Encrypted peer-to-peer sync of indexes across a user's own devices.
- Admin dashboard with anonymous, content-free health metrics.
- Fine-tuned domain adapters (legal, medical) that run locally.

15. Conclusion

EdgeMind demonstrates that useful, private AI does not need the cloud when the hardware is designed for it. By running models on the Snapdragon NPU, building ARM64-native, and adapting to power state, it turns the strengths of Snapdragon-powered HP PCs into an everyday capability: an assistant that knows a user's own material, respects its confidentiality and lasts through the workday.

The accompanying repository contains a tested prototype of the orchestration, policy and retrieval layers. The next step is validating the NPU path on device and completing the speech and language integrations outlined in the roadmap.

Appendix A: Prototype and repository

The accompanying Git repository contains **edgemind.py**, a single-file reference implementation, plus tests and sample data.

Item	Detail
Repository	github.com/ayush2459/edgemind-snapdragon (to be created from the provided folder)
Main file	edgemind.py: hardware detection, policy engine, ONNX Runtime session builder, embedders, vector store, Q&A, summariser, CLI
Commands	info, ingest, ask, summarize, bench
Tests	7 pytest tests covering policy tiers, chunking, embeddings, storage round trip and summarisation (all passing on CPU)
Not yet verified	NPU execution and benchmarks on Snapdragon hardware; LLM generation path

Appendix B: Glossary

Term	Meaning
NPU	Neural Processing Unit, a chip block built for efficient AI inference. On Snapdragon X it is the Hexagon NPU.
QNN EP	Qualcomm AI Engine Direct execution provider for ONNX Runtime; runs models on the Hexagon NPU (HTP backend).
ONNX Runtime	Cross-platform inference engine for models in ONNX format.
RAG	Retrieval-augmented generation: answers are generated from retrieved source passages.
Quantisation	Reducing numeric precision of model weights (for example INT8 or 4-bit) to save memory and speed up inference.
ARM64	The 64-bit ARM instruction set used by Snapdragon X CPUs; native builds avoid emulation overhead.
TOPS	Trillions of operations per second, a peak throughput measure for AI accelerators.